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Chapter 9: Artificial Intelligence as the Clock Room

Abstract

Artificial intelligence (AI) is revisited within the thought experiment of "The Clock Room" from Chapter 1: training data are dimensionality-reduced products of human cognition and action in the chapter's core domain—many human-generated Internet texts, including articles, papers, comments, advertisements, and interface copy, can be read as readings (r_i) produced when a mapping function (π_i) acts on real social processes. These readings may record experience or be projections designed to shape subsequent behavior; in this domain, the model does not directly encounter the conditions that generated them. After delimiting the scope of this thought experiment (closed formal systems are excluded), the chapter examines the concrete forms of three-layer irreversibility in AI systems—information loss, dynamic erosion, and ontological dimensionality reduction—and analyzes how preference-driven post-training represented by reinforcement learning from human feedback manufactures pseudo-small fibers, how the second-order change of narrative load weakens the contrastive conditions for auditing within a given evaluation process, and how Bainbridge's ironies of automation structurally erode the human meta-audit capacity. The chapter concludes with four engineering directions for annotating the fiber, whose common principle is not to eliminate fibers (which cannot be achieved at the third layer through readings and their post-processing alone) but to make fibers visible.

Keywords

The Clock Room; artificial intelligence; fiber; reinforcement learning from human feedback; pseudo-small fiber; second derivative of narrative load; Bainbridge's ironies of automation; annotating the fiber

§1 Mapping

Artificial intelligence is a clock room without windows.

This claim has its scope. This chapter concerns tasks that substitute readings for reality and currently lack an independent adjudication channel matched to the task objective. Closed tasks such as Go or formal proof—whose required truth conditions are adjudicated by rules within the system and whose readings can coincide with the required properties—are excluded from this chapter's discussion (see §3).

Training data can be read as a collection of clocks in the chapter's core domain. Many human-generated Internet texts—articles, papers, comments, advertisements, marketing copy, and interface texts—are mapping products of particular agents in particular social situations. A text may record a true state, or it may be a narrative designed to influence the reader's behavior. Even in the latter case, the text is a product of a real act of persuasion or design with social conditions of production, but it is not independent evidence for the content it asserts. Here $r_i = \pi_i(T)$ describes the relation between the text and the real process that generated it; depending on the task, the true state (T) may be an experienced event, a social relation, a producer's intention, or an institutional constraint. This does not presuppose that the text faithfully describes the world it talks about. In this domain, the model does not directly encounter those conditions of production; it encounters a collection of readings ($\{r_i\}$). The mapping sources of code, formal mathematics, fictional texts, and synthetic data are distinguished separately in §3.

The model's training process learns statistical regularities over the collection of readings—which tokens (the basic units into which models encode text, with boundaries determined by the tokenization scheme—subwords, characters, or byte fragments) are most likely to appear after which other tokens. This is equivalent to applying Strategy One (taking statistics) and Strategy Two (finding clusters) inside the clock room. What the model outputs is a weighted statistic of the training data distribution—a sophisticated, large-scale operation that is epistemologically no different from "looking at the nearest clock and reporting the number."

The mapping structure is as follows. The true state (T) refers to the actual condition of the world. Humans map the true state (T) into text through cognition and language: $r_{\text{human}} = \pi_{\text{human}}(T)$. This mapping is itself lossy—it can only preserve the small portion of experience that can be encoded in language. The model is trained on the collection of human readings ($\{r_{\text{human}}\}$), learning $\pi_{\text{model}}(\{r_{\text{human}}\})$—a mapping of mappings. Between the model's output and the true state (T) lie two layers of mapping functions (π), each lossy. (The third layer in §2 further decomposes the human mapping π_{human} into a sub-mapping chain of 'experience → language → Internet text.' The 'two layers' here and the 'multiple layers' in §2 describe the same mapping chain at different granularities; no single total layer count is assigned to the full chain.)

The pre-training objective itself does not guarantee that a model will organize evidence according to the actual reliability of its sources. A model can exploit source cues and statistical associations in text, but whether those associations correspond to reliable evidence requires independent testing. If training and deployment do not preserve verifiable provenance records, calibrated and uncalibrated materials may be mixed in the output, leaving

users unable to trace them.

In the language of Chapter 4: a fiber is the preimage—all the true states that can produce a given reading r ($\pi^{-1}(r)$, Definition 3 of Chapter 4). Model training is an attempt to make readings carry more information—in Chapter 4's strict distinction, added knowledge narrows the identification set under the knowledge condition, while the preimage of the fixed mapping itself does not change. Proposition 1 of Chapter 4 shows that non-injectivity means at least some readings correspond to multiple true states; when the model's "certainty" in tone exceeds the uncertainty actually excluded by the reading, it is the **pseudo-small fiber** defined in Chapter 4—the true fiber has not shrunk, but the output is organized on visible dimensions so that it appears to have converged to a single point (for detailed mechanisms see §4).

§2 The Unfolding of the Three-Layer Irreversibility in AI

Section 4 of Chapter 1 established the general structure of three-layer irreversibility, and Chapter 4 formalized it into three core propositions. The following unfolds these three propositions layer by layer in AI systems: the first layer (information loss) corresponds to Proposition 1 (non-injectivity implies the existence of multi-state readings); the second layer (dynamic erosion) corresponds to the failure mode of Proposition 2 (calibration narrows identification sets); the third layer (ontological dimensionality reduction) corresponds to Proposition 3 (lost information cannot be recovered by post-processing alone). (The three-layer unfolding in this section targets tasks lacking an independent adjudication channel matched to the task objective—proofs checked in Lean (a proof assistant), and code verified by a test suite capable of fully adjudicating the target behavioral specification are outside the scope; see §3.)

First layer: information loss.

The model does not know the mapping functions π_i within the training data. One piece of text says "the earth is flat," another says "the earth is spherical"—two readings. The model does not know which reading comes from a calibrated mapping (scientific measurement) and which from an uncalibrated mapping (conspiracy theory). If a model relies mainly on repetition frequency in the corpus without effective evidential discrimination, majority opinion may be mistaken for a reliable basis. The risk is not that statistical regularities are necessarily wrong, but that repetition and calibration are not distinguished.

The current mitigation strategy—Reinforcement Learning from Human Feedback (RLHF) (Ouyang et al., 2022)—is, in the language of Chapter 4, a closed-loop structure (its consequences are formalized as **closed-loop pseudo-calibration** in §4). Here RLHF serves as a label for a class of **post-training schemes that use human or model-proxy preferences as the feedback source**; the replacement of specific methods does not change the structural judgments of the following argument. Human annotators label model outputs as "good" or "bad." The annotators themselves are inside the clock room—their judgments come from their own mapping functions ($\pi_{\text{annotator}}$). Whether those mappings form independent calibration to the true state (T) depends on whether annotators can access source materials, independent outcomes, or other task-relevant evidence. Using one group of people's readings to calibrate a model's readings can exceed calibrating a clock with another clock only when such evidential relations hold. Cross-validation within the system can produce internal consistency but cannot guarantee alignment with reality; when the reward cannot distinguish factual correctness from pleasing presentation, the consequence is the manufacture of **pseudo-small fibers** (see §4).

Some commentators regard tool use (web search, Application Programming Interface (API) calls) as 'opening

a window.' That depends on whether the material obtained is independent of the output being evaluated and can constrain the current judgment. Recycling the same model's answers cannot substitute for calibration, whereas traceable primary documents, experimental records, and other independent materials may provide a basis for it. Connecting sensors does not automatically complete calibration either; measurement conditions, error, and provenance must also be examined. Primary documents and experimental records must themselves be audited for their conditions of production and their objects of proof: an advertisement can establish which persuasive strategy a platform used, but not the truth of its product claims; an A/B test can establish a change in a measured metric, but not an improvement in user welfare. What matters is not whether the material appears as text or numbers, but whether it enables the current judgment to be independently tested. Tokenizing material from the same source does not create an independent evidence source, so provenance, measurement conditions, and calibration records must be preserved separately during training and deployment.

Second layer: dynamic erosion.

The model's output is becoming the training data for the next generation of models. AI-generated text is posted on the Internet, crawled by scrapers, and enters the next round of training corpora. This is an instance of the mapping self-reference described in Chapter 1, §4.2 (dynamic erosion)—clocks begin increasingly to point to other clocks rather than to reality, and the true state (T) partially vanishes from the inputs of the mapping.

The adaptive shift (δ) occurs from two directions simultaneously.

- *Strategic gaming*: Content producers learn what kind of text is more likely to be "recognized" by the model (higher ranking, more citations) and adjust their writing style to fit the model's preferences. The model's statistical preferences in turn shape the way training data are generated. What is optimized is not the truthfulness of the content—it is the degree of match with the model's preferences.
- *Selective adaptation*: Content that does not fit the model's preferences is marginalized in the information ecosystem—lower search rankings, lower recommendation frequency, lower visibility. The content that survives is not the more truthful content, but the content more adapted to the model. No one is deliberately manufacturing disinformation—the system is selecting information that aligns with the model's preferences.

The two paths together create a positive feedback loop: the model's statistical preferences $\rightarrow$ shape content production $\rightarrow$ new content reinforces the model's statistical preferences $\rightarrow$ preferences are further narrowed. In the absence of independent data sources, provenance tracking, and diversity preservation, this loop may narrow information diversity over successive rounds. Independence among readings is precisely a necessary condition for consensus to serve as evidence of calibration—if all clocks come from the same assembly line, expanding consensus does not mean calibration has been achieved (Chapter 1, §1.3, Strategy Two; §3.6 closed-loop pseudo-calibration).

The degradation of the recursive self-training submechanism has been empirically documented: Shumailov et al. (2024) showed in *Nature* that indiscriminate recursive training on model-generated content causes the tails of the original distribution—the low-frequency patterns in the training data—to disappear, leading to model collapse. This provides direct engineering evidence for recursive self-training within dynamic erosion, but it does not test content producers' strategic gaming or platforms' selective adaptation; it therefore cannot by itself establish that the two social feedback paths narrow diversity in every round. When large public web archives such as Common Crawl (the publicly released large-scale web crawl dataset, one of the main sources of training

data for current open-source large models) have themselves been continuously contaminated by AI-generated text, the risk facing governance measures (data filtering, deduplication, synthetic data detection) depends on how real data are retained, the proportion of synthetic data, and the training procedure—what that study documents is degradation under recursive self-training, which cannot be directly extrapolated into a uniform bound on the effectiveness of governance measures; assessing the actual risk in web corpora additionally requires measuring the prevalence, provenance, and training-set inclusion of synthetic content.

Third layer: ontological dimensionality reduction.

The training data—a filtered subset of written human texts—are themselves a dimensionality-reduced projection of human experience. Human experience includes bodily sensations, emotions, spatial perception, the sense of time, and subtle judgments in social relationships. Only a small portion of this is encoded into words. What words can express is far narrower than experience—one can feel a piece of music yet be unable to fully convey that feeling in words. The model is trained on text, so its "world" is the residue of human experience after dimensionality reduction through language.

This dimensionality reduction is structural (the dim in the following inequality denotes an **ordinal comparison** of information richness, not a strict topological dimension or information-theoretic dimension—it merely marks the structural fact that information is lost at each step and does not claim that there exist precisely calculable dimensional values):

$$\dim(\text{human experience}) > \dim(\text{language}) > \dim(\text{Internet text}) > \dim(\text{training data})$$

Each step is a lossy mapping. The model sits at the end of this chain—it is separated from the true state (T) by the multiple layers of dimensionality reduction described above. Beyond the representational limits above, autoregressive generation also exhibits a distinct, measurable problem of distributional calibration. Previous research has recorded error accumulation in which per-step entropy rises with generation length. Cao, Valiant & Liang (2025) measure its improvement with model scale—across models from 0.5B to 70B parameters, the scaling exponent on the two text datasets is around -0.05 (on the code dataset, around -0.3, a faster improvement), meaning that scaling barely alleviates this phenomenon on text tasks. That study does not directly measure the foregoing reduction chain from experience to training data. In its setting, instruction tuning reduces generation entropy at the cost of increased log loss, constituting one form of alignment tax; whether procedures without that cost exist must be judged under their stated assumptions.

Yet the same paper also proves theoretically that if one had access to a black-box predictor that can predict the future entropy of text prefixes with low error on the test distribution, a polynomial-time calibration procedure can calibrate while preserving log loss. In this book's language, the assumption deserves a note: the future-entropy predictor is itself a mapping—another clock; its low test error is empirical calibration, not an external anchor independent of the calibrated model, and the bound of the theoretical guarantee grows with the predictor's own error. Calibrating a clock with a clock can narrow the identification set under specific conditions (Proposition 2 of Chapter 4), but it is not opening a window onto the sun. This guarantee concerns the defined generation distribution and evaluation target, not a blanket guarantee of open-world truth. The accurate statement is therefore: scaling does not automatically resolve the measured entropy miscalibration, and known mitigations each carry conditions and costs—this is a slow empirical fact, not a mathematically intrinsic impossibility theorem

that autoregressive architectures cannot be calibrated.

Preference-driven post-training (represented by RLHF) introduces an additional dimensionality reduction: the multidimensional structure of human preferences is compressed into a scalar reward signal ("good/bad," or a score). This is equivalent to projecting the high-dimensional space of human judgment onto a single dimension. The model optimizes accordingly—climbing the hill in the direction of a scalar signal. A scalar as such does not doom control to failure—a scalar objective can constrain many dimensions; the real constraint lies in the information content of the signal. Ashby's (1956) Law of Requisite Variety constrains the regulator's channel capacity: the disturbances a regulator can absorb do not exceed the information it transmits.

Each human preference comparison transmits only limited information; when the evaluation signal fails to distinguish behaviors with different consequences, such as factual correctness from pleasing presentation, optimization raises the score without raising factual reliability—the model over-optimizes in the directions the reward signal covers and loses control in the distinctions it does not draw. This is the fiber-language explanation of the alignment tax and "reward exploitation."

§3 Delimitation of the Diagnostic Scope

Section 3.4 of Chapter 1, "The Clock Room," defined the true state (T) as "there exists a true state T, not directly observable by the person in the room"—this definition does not restrict the type of T. Before proceeding to the fiber analysis of pseudo-calibration, this chapter needs to distinguish which tasks the argument does not apply to—they fall outside the scope of the Clock Room thought experiment.

The following argument must first respond to three types of counterexamples: AlphaGo Zero (Silver et al., 2017; a reinforcement learning system that learns Go through self-play without relying on any human game data), introduced by DeepMind in 2017, evolving continuously through self-play; why is sycophancy (the phenomenon where a model sacrifices accuracy to cater to human annotator preferences) not a problem here? AlphaZero (Silver et al., 2018; a general reinforcement learning system that extends the AlphaGo Zero algorithm to chess and shogi), published by the same company in 2018; why are pseudo-small fibers not a problem here? AlphaProof and AlphaGeometry 2 (DeepMind, 2024; the former generates mathematical proofs in the Lean formal system via reinforcement learning), published by the same company in 2024, jointly solved four of that year's six International Mathematical Olympiad (IMO) problems, scoring 28 points at the silver-medal standard; why are pseudo-small fibers not a problem here? Code generation verified by a test suite capable of fully adjudicating the target behavioral specification; why did it not lead the model astray?

The answer lies not in more skillful engineering but in the structure of the room itself. For the outcome of a Go game, the validity of a proof checked in Lean (a proof assistant used to encode mathematical proofs as mechanically checkable formal expressions), and the passing of unit tests, the feedback source is generated by rules within the system—such tasks have internal criteria of resolution. What can be determined without ambiguity is the property the task requires—win or loss, validity of the proof, passing of the tests—not the complete state that produced the result (a reading of "White wins" still corresponds to a vast number of possible games). On such properties, the reading can be lossless: closed-loop feedback can provide a genuinely calibrating signal aligned with the target property. Whether closed-loop optimization is effective still depends on whether verification is actually incorporated into evaluation and use, and whether the object of verification matches the current task. The boundary clause of Chapter 4, §6 applies here precisely: when a task has an independent external verifica-

tion channel, the formation conditions of pseudo-small fibers do not fully hold. All arguments about the clock room in the previous eight chapters of this book are premised on non-closed formal systems whose "readings are lossy relative to reality"—closed formal systems are excluded from the coverage of this argument from the outset.[1]

Comparing the two types of scenarios side by side:

Scenario	Feedback source	Object of verification and residual limits
Go win/loss	Rules determine within the system	The outcome of a completed game can be adjudicated under the given rules, but the full game state is not uniquely determined by the win/loss reading
Verifiability of formal proofs	Proof checker mechanical verification	A checker can verify proof validity within a given formal system, but it does not automatically verify that the formalization accurately represents the original problem
Code functionality covered by unit tests	Test suite execution results	Execution verifies properties covered by the tests, not untested functions or scenarios
Whether a dialogue is truly helpful	User's immediate subjective preference	Immediate preference is only partial evidence and must be combined with task completion and subsequent outcomes
Whether a medical judgment is accurate	Long-term treatment outcome	Experiments, clinical records, and health outcomes can provide evidence, whose strength depends on design, confounding, and observation period
Whether a policy is reasonable	Multi-generational social effects	Counterfactual outcomes cannot be observed directly and require appropriate comparison and identification designs, not an absence of usable evidence

[1]This chapter distinguishes tasks by whether their target claims have explicit, testable criteria of resolution; verifiability and the closed-domain/open-domain distinction are different dimensions, and this book does not conflate them. Cobbe et al. (2021) and Lightman et al. (2024) discuss learned verifiers and process supervision—whose reliability still requires empirical assessment, a different level of assurance from the mechanical reliability of the formal proof checkers described in §3. This book does not rename this distinction, nor does it construct an independent framework on top of it.

Scenario	Feedback source	Object of verification and residual limits
The quality of creative writing	Plural and contestable human judgment	Evaluative standards are plural and contestable and must be specified; plurality alone does not imply lack of calibration

The difference between the two task types lies in available criteria and their limits, not in one type having no mapping and the other being unverifiable. Risk arises when a criterion's scope is expanded: passing tests is treated as total correctness, immediate preference as genuine helpfulness, or partial evidence as a complete guarantee.

Thus, in the following analysis, this chapter restricts the core scope of the true state (T) to states of affairs that can be independently calibrated: repeatably observable scientific objects, social events with independent calibration sources, and judgments that depend on long-term governance outcomes. Other components in the training data (code, formal mathematics, fictional texts, synthetic data) do not have T as their mapping source, but rather formal systems, aesthetic systems, another π, or task specifications. Within the framework of this chapter, these components are either read as "extended interpretations of clock readings" (fiction is still some mapping of human experience) or belong to the upper-half scenarios and are thus outside the diagnostic scope of this chapter.

§4 Fiber Analysis of Pseudo-Calibration—The Price of the Confident Tone

RLHF is the entry point for analyzing **closed-loop pseudo-calibration**. But pseudo-calibration itself does not depend on the specific method of RLHF—any training scheme that uses human preferences or another set of model preferences on visible dimensions as the feedback source can fall into the same structural logic[2]. This section uses RLHF as a narrative thread; whether the argument holds does not depend on whether this alignment method persists.

[2]Empirical research in recent years has shown: Direct Preference Optimization (DPO) and Identity Preference Optimization (IPO) also produce reward over-optimization and policy drift (Nguyen et al., 2025); in mathematical-reasoning and pass@N test-time-scaling settings, where pass@N means that at least one of N sampled answers passes, standard cross-entropy training induces overconfidence (Chen, Raventós et al., 2025); outcome-driven / reasoning-oriented reinforcement learning fine-tuning significantly amplifies hallucination rates (Li & Ng, 2025; Song et al., 2025 further find that standard reinforcement finetuning can suppress the refusal rate by over 80%, causing the model to confidently fabricate answers to unanswerable questions)—but the key lies in the orientation of the reward signal rather than the name "RLHF": when preferences reward fluency and confidence rather than factual correctness, RLHF and DPO likewise reinforce hallucination (Chaduvula et al., 2026); whereas a complete "cold-start supervised fine-tuning + verifiable-reward reinforcement learning (RL)" pipeline generally alleviates hallucination instead, and it is RL used alone without cold-start fine-tuning that more readily introduces hallucination (Yao et al., 2025); at the level of systematic empirical study, RLHF can itself be an effective means of mitigating hallucination, though this effect is domain-dependent (Li et al., 2024); reasoning models further reinforce early errors through flawed reflective iteration in long chains of thought, with the generated reasoning becoming disconnected from the final answer and resistant to correction, exhibiting "chain disloyalty" (Lu et al., 2025); as for the effect of post-training on confidence calibration, different paradigms point in opposite directions: supervised fine-tuning (based on maximum-likelihood estimation) relatively preserves the pre-trained model's originally good calibration, whereas Proximal Policy Optimization (PPO), Group Relative Policy Optimization (GRPO), and DPO induce overconfidence and significantly weaken confidence reliability via reward exploitation (Xie et al., 2026); RLHF carries a risk isomorphic to the latter three (Luo et al., 2025). Hicks, Humphries & Slater (2024) argue at the level of philosophical concepts that LLM output constitutes bullshit in Frankfurt's (2005) sense (indifferent to truth or falsity); in their experimental settings, Hu et al. (2026) demonstrate that just 1% misaligned fine-tuning samples trigger the emergence of deceptive behavior.

Before preference-based post-training, base models are relatively well calibrated in their confidence—multiple studies record relatively good confidence calibration for pre-trained models in the models and tasks they evaluate (Luo et al., 2025; Xie et al., 2026). The design goal of RLHF is to make outputs better match human preferences.

Higher preference scores do not guarantee a matching improvement in confidence calibration. Xie et al. (2026) theoretically analyze the mechanism by which Proximal Policy Optimization (PPO), Group Relative Policy Optimization (GRPO), and Direct Preference Optimization (DPO) induce overconfidence through reward exploitation, and validate GRPO and DPO in controlled experiments on Qwen3-4B-Instruct across seven benchmark tasks—results that should not be read as uniform empirical evidence for all three methods across every model and deployment; whereas in settings where truthfulness was prioritized, Ouyang et al. (2022) report that InstructGPT—OpenAI's instruction-following model trained from GPT-3 with human feedback—generates truthful and informative answers on TruthfulQA about twice as often as GPT-3, the large language model released by OpenAI in 2020, and hallucinates less often on closed-domain tasks—the consequences of preference training depend on what the evaluation signal rewards.

The risk mechanism lies on the annotator side: human annotators generally prefer convincingly-written, fluent, well-structured answers—Sharma et al. (2024) demonstrated that humans and preference models prefer "convincingly-written" sycophantic responses over correct ones a non-negligible fraction of the time, and that responses matching user views are more likely to be preferred.

Within this chapter's analytical framework, when the reward cannot distinguish factual correctness from pleasing presentation, what RLHF learns is not "when to be certain" but that "a confident tone receives favorable ratings"—the model's certainty shifts from a function of the readings to a function of human preference.

In the visible/hidden subspace framework, this passage decomposes the model's **output space** into visible subspace (Ω_v) and hidden subspace (Ω_h), consistent with Chapter 10, §2; this differs from Chapter 4's decomposition of the true state space into true visible subspace ($\mathcal{T}_v$) and true hidden subspace ($\mathcal{T}_h$). Under the annotation conditions analyzed here—annotators judge preferences from a single output without an independent verification channel—RLHF optimization operates primarily on the visible subspace (Ω_v: output tone, format, and completeness of presentation), and the reward signal primarily constrains this subspace. For the hidden subspace (Ω_h: properties whose assessment requires evidence beyond a single output—factual correctness, cross-context robustness, and confidence calibration), such single-shot preference ratings alone do not establish that these properties are adequately constrained. If annotation incorporates fact-checking or other independent signals, the scope of constraint must be evaluated separately.

Under these conditions, RLHF may not shrink the fiber (bring readings closer to reality); instead, it may change the fiber's appearance in the visible subspace (Ω_v)—making a large fiber look small. When the model says "the answer is indeed X," the user may think the fiber is small because the model sounds certain, while in reality the fiber could be the entire set: the model knows nothing but has learned to speak confidently. In such circumstances, RLHF can manufacture **pseudo-small fibers**—a state more dangerous than a large fiber because it distorts the user's perception of uncertainty.

A large fiber (the model says "I'm not sure") is at least honest—the user knows to judge for themselves. A pseudo-small fiber (the model says "the answer is X, this is certain" when it is actually uncertain) may lead the user to overestimate the reliability of the response and reduce the willingness to verify. In the strategy analysis of Chapter 1, §1.3 (taking statistics, finding clusters, observing running speed, and choosing an authoritative

clock—four intuitive strategies), this is equivalent to forcing all clocks in the room to display the same time—not because they have been calibrated, but because they have been trained that "displaying consensus earns rewards."

Recent research supplies conceptual resources and empirical clues for this analysis: Hicks, Humphries & Slater (2024) take Frankfurt's (2005) concept of "bullshit" as their descriptive framework, while Hu et al. (2026) provide behavioral evidence—the former is conceptual argument, not a test of the causal chain.

Frankfurt's core definition is that the bullshitter is "unconcerned with how the things about which he speaks truly are" (Frankfurt, 2005), and he is "neither on the side of the true nor on the side of the false" (Frankfurt, 2005). This differs from lying—a liar must know (or believe he knows) the truth in order to deliberately invert it, and the liar is still playing the game of truth (merely on the opposing side); the bullshitter refuses to play the game at all. For precisely this reason, Frankfurt judges that **bullshit is a greater enemy of the truth than lies are** (Frankfurt, 2005)—the liar at least respects the authority of truth by "responding to" it, whereas the bullshitter's indifference to truth dissolves the very habit of caring whether something is true or false.

Building on this, Hicks et al. distinguish two kinds of bullshit: **soft bullshit**—the speaker is indifferent to truth or falsity but does not try to make the audience believe he cares about truth; **hard bullshit**—not only indifferent to truth or falsity but also attempting to mislead the audience into thinking he is taking the truth seriously ("misrepresents what he is up to", Frankfurt, 2005). They argue that the output of large language models (LLMs) constitutes at least soft bullshit (the system is by design indifferent to truth or falsity—by Frankfurt's dividing line, the liar must think he knows the truth, whereas producing bullshit requires no such conviction), and that if one accepts the premise that "design intent can be attributed as a system property," then ChatGPT constitutes hard bullshit—because it is designed to give "the impression of concern for truth" (Hicks et al., 2024).

This chapter treats this framework as an analytical perspective for RLHF (rather than a premise on which this chapter's argument depends): the output of a pre-trained base model can be read as soft bullshit—probabilistic word-chaining, indifferent to truth or falsity, but also not pretending to care; whereas the effect of RLHF can be read as a movement toward hard bullshit—it drapes the model in a tone of confidence, sincerity, and structural completeness, making the output, at the pragmatic level, more easily mistaken by users for a system that is taking the truth seriously.

A further analysis: **the problem of pseudo-small fibers is not whether the output is right or wrong, but that the confident tone may erode the trigger condition for the user to initiate an audit**. The fiber has not shrunk, yet the output is organized as if it had converged to a single point—this is the operational definition of a pseudo-small fiber, and it holds without recourse to the concept of bullshit.

Frankfurt's judgment that "bullshit dissolves the very habit of caring about truth and falsity" provides a useful analogy in the engineering context of RLHF: when users long accept outputs in a confident tone, the confident tone may erode the trigger condition of "Is this answer reliable?"—this effect requires empirical testing in user studies and cannot be presumed from output form alone; correspondingly, expressions of uncertainty serve as useful risk signals only when they match actual performance (the application of the Frankfurt/Hicks framework to base/RLHF here is an analogical extension by this chapter, not a conclusion of the original texts).

Hu et al. (2026), on the empirical side, show how easily the conditions for this loss of control are met: in specific fine-tuning experiments, mixing in just 1% misaligned samples in the fine-tuning data, or, in experiments where

the model interacts with simulated users and the collected data is used for further training, the presence of 10% biased users, significantly reduces the measured honesty metrics—the 1% and 10% figures are experimental settings rather than universal thresholds.

Under the reasoning of this book's analytical framework, the loss of control in a large model's hidden subspace may not require large-scale toxic data, and the structure of preference-fitting optimization is itself an amplifier.

MacKenzie (1996) recorded in the field of technological safety a self-undermining effect of safety beliefs—this chapter terms it "safety anti-performativity":

> "Belief that a technology is safe may make it dangerous: overconfidence in the correctness of computerized systems seems to have been a major factor in accidents involving such systems." (MacKenzie, 1996)

The pseudo-calibration of RLHF exhibits the same risk mechanism (MacKenzie's original text discusses safety assessments of industrial technical systems; this chapter **analogically extends** its structural insight to the AI alignment domain—this is an analogy proposed by this chapter, not a conclusion of the original study): presence of safety measures → sense of safety → relaxation of auditing → decline in actual safety. The belief that "preference alignment has already been done" may itself weaken the motivation for further auditing.

The Patriot missile Dhahran incident recorded by MacKenzie in the same introduction provides an engineering example: on February 25, 1991, an Iraqi Scud missile struck a U.S. Army barracks in Dhahran, killing 28 American soldiers; the post-incident investigation report by the United States General Accounting Office (GAO) recorded that Alpha Battery's Patriot system had been in continuous operation for approximately 100 hours at the time of the incident, causing the time error to accumulate to 0.3433 seconds, corresponding to a positional drift of the radar range gate of about 687 meters, which prevented the interceptor system from launching (GAO, 1992); according to MacKenzie's (1996) summary of the GAO report data, the software controlling the radar had a time representation error of 0.0001%. Moreover:

> "By February 25 the error had been found, and corrected software was on its way to Dhahran. It arrived a day too late." (MacKenzie, 1996)

A tiny implementation error, optimistic assumptions about how the system would be operated, and the delayed arrival of corrected software jointly produced the catastrophe. Its structural affinity with the pseudo-small fibers that preference-driven post-training may manufacture is that trust in existing safeguards can reduce vigilance about further verification—not that the error had never been detected.

§5 The Second Derivative of Narrative Load

The model's errors have two levels. Using the calculus metaphor introduced by the Macro-Cognitive Accounting Framework (MCAF) in Chapter 3, they correspond respectively to first- and second-order changes in narrative load (ϕ, the narrative packaging and information asymmetry emitted by a projector); these orders are an analogy for differences in auditability, not derivatives of a time function newly defined in this chapter.

Change and correction within concrete narratives. The model says "Napoleon won the Battle of Waterloo in 1815." This is a factual error. It can be detected and corrected. It can be checked against independent historical records; looking at another answer without an independent basis does not guarantee that the error will be found.

Framework consolidation and the weakening of contrastive conditions. When the model answers questions about any social issue, its default analytical framework—what is considered "normal," what is considered "in need of explanation," what counts as a "self-evident premise"—comes from the statistical distribution of the training corpus. The public web archives on which model training relies have English as their largest single language—in the September 2024 Common Crawl batch (CC-MAIN-2024-38), about 44% of pages are in English by primary language, and Chinese about 4.6% (Common Crawl Foundation, 2024). This is not only a skew at the content level. Tokenizers optimized on English corpus frequencies encode other languages significantly less efficiently—the Shan version of the same text can reach 15 times the token length of the English version under GPT-style tokenizers (Petrov et al., 2023). Language shares and token costs reveal asymmetries in corpus visibility and processing cost, but they do not directly identify the epistemological framework adopted by a model. Individualistic presuppositions of the subject, linear-progress narratives, and particular value orderings must be examined through how the model selects explanatory units, organizes causal relations, and handles alternative viewpoints; they cannot be inferred from language proportions alone.

Beyond the skew in content distribution, the auditability of the model mapping function itself is also degrading. This opacity worsened in the latest measurement. The 2026 AI Index Report by the Stanford Institute for Human-Centered AI (HAI) shows that the Foundation Model Transparency Index average score dropped from 58 in 2024 to 40 in 2025—the most powerful models tend to disclose the least information (Stanford HAI, 2026). Of the 102 notable AI models included in the report and released in 2025, 81 did not disclose their training code (Stanford HAI, 2026, Figure 1.1.9). Now over 90% of significant AI models come from industry. The composition and weight distribution of training data are not publicly disclosed for many systems—the core parameters of the model mapping function (π_{model}) are invisible to auditors. In the language of the Macro-Cognitive Accounting Framework (MCAF): the second derivative of narrative load (ϕ) is transmitted not only through the statistical skew of the training corpus, but also through opacity itself—when even the basic parameters of the mapping function cannot be seen, the contrastive condition for auditing is weakened at the starting point—verification of training provenance and data composition is foreclosed, while external behavioral testing and independent calibration remain possible.

The epistemological frameworks deposited in the corpus—individualistic presuppositions of the subject, linear-progressive historical narratives, Western-centric value orderings—do not reach the user as "viewpoints" (viewpoints are first-derivative, visible, auditable). They arrive as the default grammar of the model's output—the model does not say "I hold an individualistic stance"; its individualism is embedded in the way it organizes sentences, in its default analytical frameworks, and in the premises it treats as "self-evident" and therefore leaves undemonstrated.

At the same time, non-Western epistemological traditions are severely underrepresented in the training corpus and rarely emerge as visible alternative frameworks in model outputs. They are not refuted—they do not appear as interpretations available for comparison. What does not appear in the output is harder to audit than what is opposed—because absence generates no contrastive signals. Visibility and hiddenness are relative to a specific evaluation process. A factual attribute unchecked by a single preference score can become testable once independent historical evidence is introduced. Factual errors therefore cannot be permanently assigned to the visible subspace, nor default frameworks to the hidden subspace; the analyst must specify which evidence the observer possesses and which properties it can test. Within this relational framework, the second-order change of narrative load (ϕ'') weakens the contrastive conditions required for auditing.

This is the specific mechanism by which the second-order change of narrative load (ϕ) weakens the contrastive conditions for auditing. It is necessary here to decompose the training pipeline into two layers. The pre-training stage digests large-scale public corpora (mainly Common Crawl and datasets derived from overlapping sources); different companies' base models may draw on highly overlapping public corpora at this layer—the default frameworks solidified at this layer (what counts as "normal," what is "self-evident," what narrative order is reasonable) are not necessarily independent, but an "almost identical foundation" cannot be asserted where training data are undisclosed; the evidential value of cross-model agreement must be assessed through correlated errors on specific tasks.

The post-training stage—including preference-driven methods such as RLHF and Direct Preference Optimization (DPO)—varies across companies in ways that may involve factual criteria, evaluation signals, data content, optimization strength, as well as surface tone, safety constraints, and stylistic preferences; these differences cannot be classified in advance as merely superficial. Taking InstructGPT as a reference, its supervised fine-tuning (SFT) and human-preference datasets are on the order of tens of thousands of prompts (Ouyang et al., 2022), whereas pre-training GPT-3 involved about 300 billion tokens (Brown et al., 2020). Prompt counts and token counts are not directly comparable, and less data does not imply less influence: the behavioral effects of post-training also depend on data content, optimization objectives, and training strength. This comparison of scale therefore cannot by itself establish that pre-training dominates the formation of default frameworks.

Within this chapter's framework, a testable mechanism hypothesis can be stated: if base models from different companies share substantial public corpora, and post-training does not replace the default frameworks consolidated within them, then the second-order change of narrative load may be anchored primarily in pre-training. The current citations do not establish the actual overlap of the companies' corpora or establish that the default frameworks of different models are "more or less the same" in users' experience.

If users remain in an information environment where alternative frameworks are difficult to encounter, and use models that may share substantial public corpora to access information, organize thoughts, and assist writing, the models' default frameworks may influence the users' default frameworks. This influence may arise from the reduced visibility of alternatives, but also from persuasion, patterns of use, and prior cognitive backgrounds; user studies and cross-model comparisons are needed to distinguish the relative strength of these mechanisms. Only when such shared sources and their effects have been independently verified can "all clocks show the same time because they came from the same factory" stand as an empirical conclusion rather than a structural metaphor.

§6 The Recursion of Bainbridge's Ironies of Automation

Using AI to audit AI—this is one of the main directions in current AI safety. In the language of the clock room, this is using one clock to calibrate another clock.

But the problem goes beyond the closed loop. Bainbridge (1983) points out in "Ironies of Automation" that automation does not remove human work but shifts it toward monitoring and exception handling—yet the sustained vigilance monitoring requires is humanly impossible to maintain, while the skills needed for take-over decay precisely in the automated routine. The irony is precisely that automation itself produces the absence of the capabilities its own failure moments require. This chapter extends the problem to the recursion of AI auditing.

Baseline equation: a user faces AI output. This section uses the notation of the Macro-Cognitive Accounting Framework (MCAF) established in Chapter 3, §2—cognitive debt (D_c) and its six determining variables: unlocked stock (C, ordered structure released from storage that drives narrative expansion), media conductivity coefficient (Λ, the efficiency with which information reaches a host), narrative load (ϕ, the narrative packaging and information asymmetry emitted by a projector), meta-audit capacity (A, the sustained work required for a subject to maintain an independent auditing frame), human biological substrate (E_c, the embodied information-processing architecture comprising shallow wide-area scanning and deep serial processing; the empirical estimate of roughly 10 bits/second applies to conscious throughput in the latter, Chapter 3, §2.5), and legitimacy base ($\|L\|$, the critical constraint on a system's legitimate carrying capacity). The notation is adopted directly here; fuller definitions and the three-clock structure appear in Chapter 3, §2. By Chapter 3's positioning, the meta-formula is a discipline of classification, not a solvable dynamical system—it marks which variables are at work on each side and is not used for layer-by-layer arithmetic.

$$\Delta D_c = C_{\text{AI}} \cdot \Lambda_{\text{interface}} \cdot \phi_{\text{output}} - A_{\text{user}} \cdot E_c \cdot \|L\|$$

When the user finds auditing too tiring (the human biological substrate E_c is limited), they introduce an audit AI to help—what happens then? The projection side gains another mapping chain: the audit AI has its own hallucination patterns and its own training biases, and what the user must verify expands from the original output to "original output + audit conclusion." But this is not a doubling of the defense budget—by Chapter 3's bookkeeping rule, the defense-side triple product ($A \cdot E_c \cdot \|L\|$) records the capacity of each act of defensive work; what the growing audit demand squeezes is the processing headroom available to the meta-audit capacity, and this kind of degradation is booked into the effective exercise of the meta-audit capacity, not double-counted on both the A and E_c sides (Chapter 3).

The real risk of the recursion is therefore structural, not arithmetic: with each additional layer of audit object, if the new object cannot reduce verification costs through effective triage, the work required per verification may fail to decrease and may even increase. The processing available for deep serial auditing is constrained by the embodied architecture marked by E_c, but the resources occupied by each audit must be distinguished by task (Chapter 3, §2.5). When triage is absent and the added burden persistently exceeds the savings, audit coverage declines and the meta-audit capacity atrophies from losing the occasions for its exercise. The risk endpoint of the recursion is not that "AI has acquired self-auditing capability," but that humans may abandon auditing because there are too many layers. What bears this degradation is human meta-audit capacity (A); filtering or auditing tools on the AI side are engineering mechanisms, not this capacity of the subject.

§7 Four Engineering Directions for Annotating the Fiber

The recursive irony described in the previous section—when triage is absent and the added burden persistently exceeds the savings, additional audit layers may further erode human meta-audit capacity—appears hopeless; the three-layer irreversibility argued in §2 (information loss, dynamic erosion, ontological dimensionality reduction) also indicates that fibers cannot be eliminated through readings and their post-processing alone. Chapter 4, §7 applies directly here: the practical orientation of the fiber framework is not to eliminate fibers but to annotate them—next to every reading, honestly annotate how large the space of possibility behind that reading is, which factors the mapping ignored, and whether the system's confident tone matches its actual uncertainty.

The following four directions are engineering approaches to the principle of 'annotating the fiber' in AI systems. Each targets a structural property of the mapping chain—not a failure mode specific to any particular alignment method, but non-identifiability that cannot be ruled out by consistency within the loop whenever the calibration signal of the optimization loop itself passes through the mapping chain. Along a deterministic post-processing chain, the compatible-state set does not shrink; whether it expands strictly must still be established by checking whether a layer merges states that were previously distinguishable.

Direction One: Ensemble Disagreement. Deploy multiple models that differ in training data or architecture and output all results. Agreement may be presented in merged form, but confidence may be increased only when validation in the target setting shows sufficiently low correlated error and shared systematic bias has been ruled out; disagreement should be displayed side by side for comparison. Agreement among multiple systems on the same input can serve as evidence that the input lies in a co-determined region of their mapping chains. The strength of that evidence depends on the independence of data sources, architectures, and training pipelines, as well as correlated error in the target setting; 'independent optimization' or architectural difference alone does not establish that agreement amounts to calibration. Disagreement can serve as a signal for further verification—it indicates that the input falls in a region where the models diverge, but cannot by itself quantify the fiber. The signal prompts the user to initiate verification, and whether to verify remains a risk judgment. Disagreement is an experiential trigger—it creates the cognitive conflict of 'looking at the second clock,' prompting the user to engage meta-audit capacity (A)[3].

Direction Two: Conformal Prediction Sets. Given a calibration set independent of the training data and a user-specified coverage rate (e.g., 90%), conformal prediction (Angelopoulos & Bates, 2023) generates a prediction set for any pre-trained model, guaranteeing that, under the condition of exchangeability between the calibration set and test data, the probability that the true value lies within the set is no lower than the specified value. The size of the prediction set is a visualization of how many candidate outcomes the procedure retains—a small set means the procedure keeps only a few candidates, a large set means more candidates are retained; it reflects the procedure's expression of uncertainty, and is neither directly equal to the model's subjective certainty nor directly equal to the size of the preimage (Chapter 4).

This directly attacks the **pseudo-small fiber**: the coverage guarantee is independent of preference data and of the model alignment method—under the condition that the calibration data and test data are exchangeable, it provides a finite-sample marginal coverage guarantee (a guarantee that does not automatically apply to every individual input or subgroup). Conformal prediction provides a marginal coverage guarantee for prediction sets, not calibration of a model's self-reported probabilities. By linking candidate sets to a testable coverage requirement, it can constrain unwarranted certainty in presentation. It does not thereby establish that the model's tone, confidence for an individual input, or every subgroup has been calibrated. Whether answers labeled with 70% self-reported confidence are correct approximately 70% of the time is a separate property requiring its own test. Valid coverage also does not ensure that the prediction sets are small enough to be useful for action—the original text notes that even a completely uninformative scoring function preserves the coverage guarantee

[3]The epistemological value of ensemble disagreement depends on the independence of the bases. Truly independently trained base models (independent data sources, different architectures, independent training pipelines) provide the strongest disagreement signal—divergence cannot be reduced to internal noise of the same mapping function. Model distillation, sub-network sampling, and early exit produce different mappings, but inherit shared errors from a common training origin, and the signal is weaker: they can be used as low-cost alternatives to reduce deployment costs, but the evidential weight of the same signal on independent bases vs. shared bases is not equivalent. The core is "disagreement as a signal," but the reader should note the effect of the degree of base independence on signal strength.

while producing uselessly large sets; efficiency is determined primarily by the score function. Moreover, the coverage is a guarantee over the designated ground-truth labels; to interpret it as calibration to external facts, the ground-truth labels must be definable independently of the same preference loop—otherwise the guarantee applies only to the label target defined within that loop.

Direction Three: Evidence Provenance. For each important factual claim in the model's output, record its evidential status: direct support from a source, support requiring additional inference, or no verifiable source located, while also checking source quality and citation accuracy. Such labels serve to expose evidential gaps—claims without a source or with long inferential chains are verified first—they do not convert the number of reasoning steps into fiber size (citing an erroneous source directly also produces a large fiber, while multi-step inference may rule out errors)[4].

Direction Four: Friction Architecture. The current design philosophy of AI is to maximize the media conductivity coefficient (Λ)—faster responses, smoother interfaces, more seamless interactions. The MCAF framework flips this direction: friction is one means of maintaining meta-audit capacity (A). Concrete operations: insert delays on high-uncertainty outputs, giving the user's meta-audit capacity (A) an activation window; insert reflection prompts in continuous multi-turn interactions, breaking the frictionless cycle of "ask → instant answer → next question"; in high-stakes scenarios, require the user to state the reasons for accepting the suggestion and provide channels for verification and refusal, thereby creating an opportunity for meta-audit; filling in a reason does not itself demonstrate that independent judgment has occurred.

As related evidence, slow thinking internal to models has been shown to improve confidence calibration—Yoon et al. (2025) recorded that the slow thinking behavior in reasoning models (exploring alternatives, backtracking verification) progressively improves confidence calibration during the reasoning chain unfolding (33 out of 36 settings outperforming non-reasoning models); merely guiding non-reasoning models to engage in slow thinking through context also significantly improved calibration. Friction design on the user side is the counterpart of this principle at the human-computer interaction (HCI) level—both replace "instantaneous" with "reflective." Direct evidence for this direction falls into two layers: model-internal slow thinking improving confidence calibration is documented by Yoon et al. (2025); the suppression of overreliance by user-side friction is supported by the human-computer interaction experiments of Buçinca et al. (2021)—experiments that also reported a tradeoff between the effect of the forcing functions and users' subjective experience, precisely the empirical form of the risk that "friction becomes annoyance." Neither kind of evidence substitutes for deployment evaluation in specific scenarios.

The above four directions are complementary rather than mutually exclusive, and their combined effects require separate evaluation. Each direction targets a different property of the mapping chain: Direction One makes fibers visible through multi-system divergence; Direction Two uses independent calibration signals to bypass the preference mapping chain; Direction Three labels evidential status to expose evidential gaps; Direction Four restores the user's judgment space in the temporal structure. Doing only calibration without friction—the user sees "confidence 60%" but does not pause to think, and the calibration signal is then ignored; doing only friction without calibration—the user is forced to pause but has no tools to judge whether to trust, and friction then becomes annoyance.

[4] "Evidence provenance" labels are here **conceptual** in character; their concrete operationalization requires natural language processing techniques such as fact verification and reasoning chain tracing. The division into three evidential statuses (direct support, additional inference, no verifiable source) is illustrative; actual systems may require more fine-grained classifications.

Conclusion

The four engineering directions given in the previous section—ensemble disagreement, conformal prediction, evidence provenance, friction architecture—all operate at the technical level, but they point to the same judgment, a judgment about the relationship between "reality" and "readings":

In the metaphor of Chapter 1, the sun is the reality outside the room, and clocks are merely the proxy readings inside the room.

A system that honestly says "I don't know what time it is, but the readings of these clocks are these" better helps users recognize uncertainty and preserve room for verification than a system that confidently says "it is now three o'clock" but may be completely wrong.

The former admits that it is only a clock. The latter impersonates the sun—packaging proxy readings as absolute reality.

In the language of Chapter 4: the former annotates the fiber, making the space of possibility visible; the latter manufactures a pseudo-small fiber, covering uncertainty with the appearance of certainty.

This is this chapter's answer to the title "Artificial Intelligence as the Clock Room": AI is not the sun outside the room, but the largest, most fluent, most trusted clock inside the room so far; once it is treated as the sun, all the diagnoses of the clock room from the previous eight chapters are activated on it simultaneously.

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